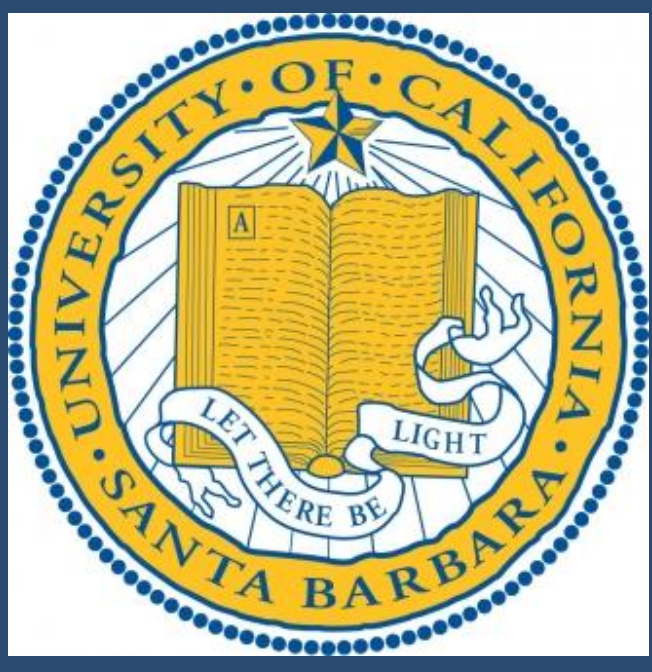
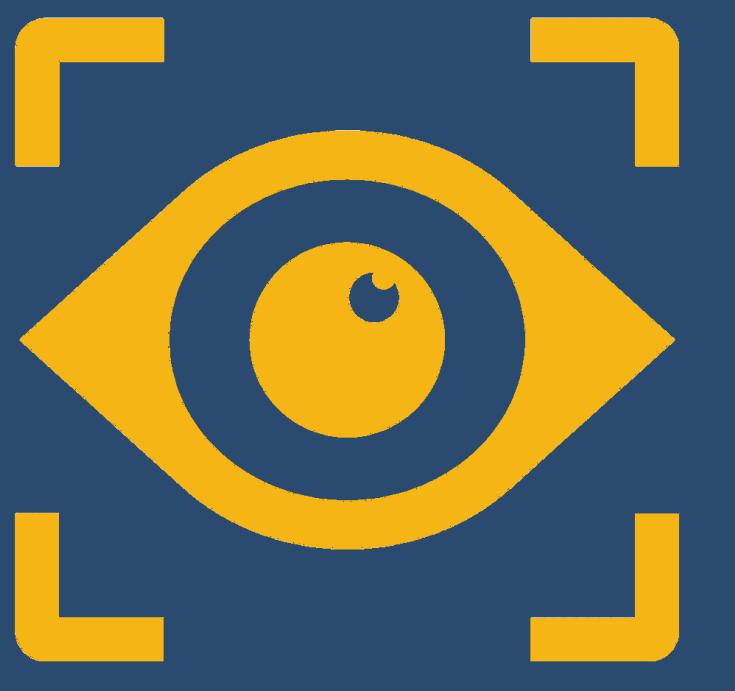




EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion

Satyam Awasthi, Vivian Ross, Tobias Höllerer, Michael Beyeler
University of California, Santa Barbara



MOTIVATION

- Eye tracking is an increasingly popular method for interacting with augmented reality (AR) and measuring user attention^{1,2}.
- Evaluating eye tracking across multiple platforms and use cases is challenging for the lack of standardized metrics.
- Existing calibration methods and accuracy measurements often do not account for the common scenarios of walking and scanning in mobile AR settings^{3,4,5}.
- We aim to elucidate differences in eye tracking performance in AR due to device types, calibration methods, and stimulus movement.



Fig.: KAT WALK C 2+ a VR Treadmill by KatVR

USER STUDY

SOLUTION

- We developed EyeTTS, an eye tracking test suite specifically designed for scenarios involving head movement and locomotion in AR.
- Test and compare different eye tracking devices on various AR tasks and metrics
- Includes tasks that require participants to maintain their gaze on stimuli, which is common in eye gaze studies^{6,7}.
- EyeTTS was created using Unity 2019.4.30f1

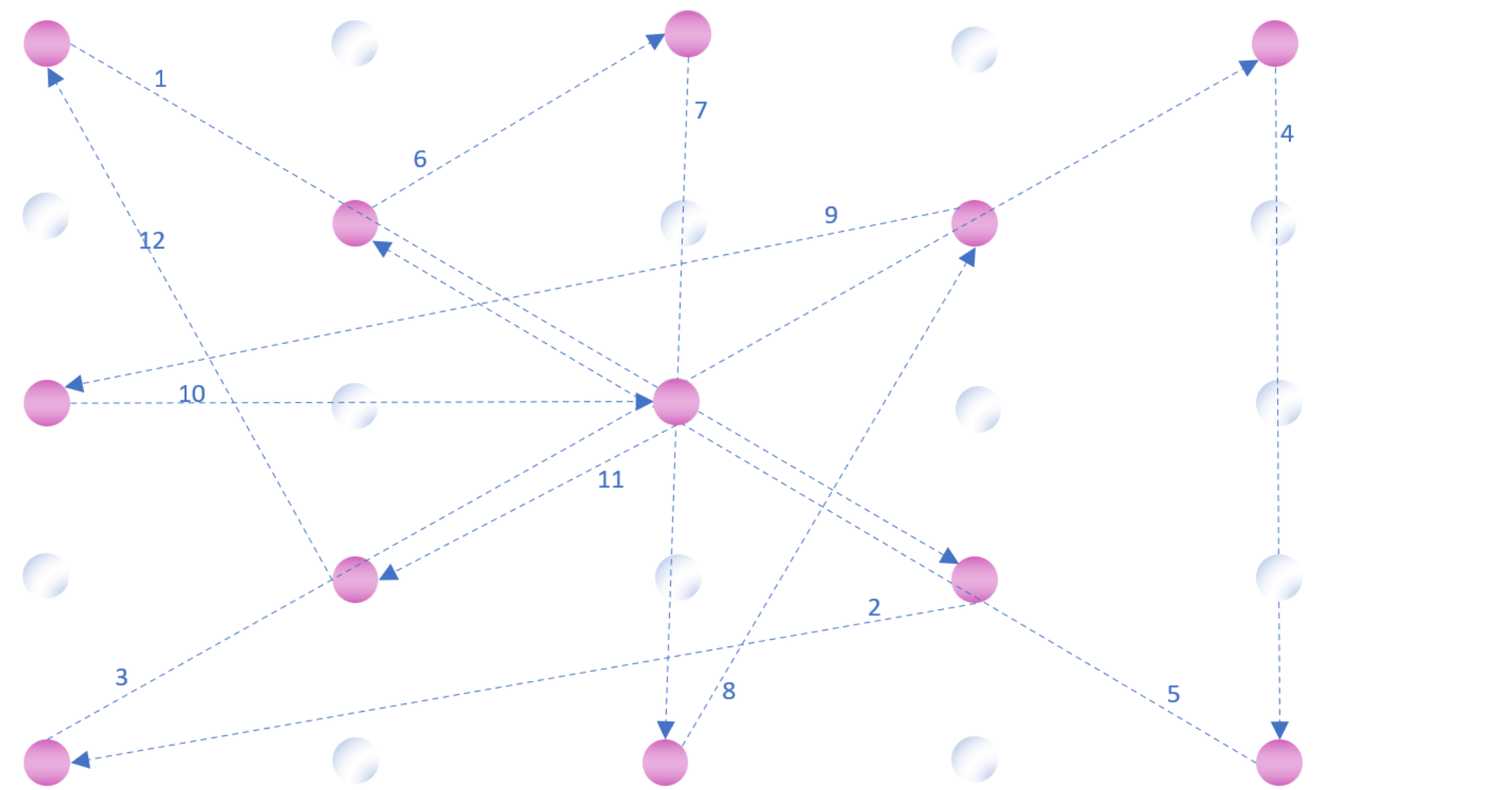


Figure 1 Stimulus moved between a grid of 5x5 on a plane. The paths it took were not obvious or known to the user.

DATA ANALYSIS

Cosine Error

The gaze vector v and the target vector u were used to calculate cosine error using the following formula:

$$\theta = \arccos \frac{\vec{u} \cdot \vec{v}}{||u|| ||v||}$$

Euclidean Error

To convert the 3D gaze direction and target vectors $v = (v_x, v_y, v_z)$ into 2D screen coordinates, we used the following formulas:

$$x = \arctan\left(\frac{v_x}{v_z}\right), \quad y = \arctan\left(\frac{v_y}{v_z}\right)$$

We then used the 2D screen coordinates to calculate Euclidean error:

$$\theta = \sqrt{(x_{target} - x_{gaze})^2 + (y_{target} - y_{gaze})^2}$$

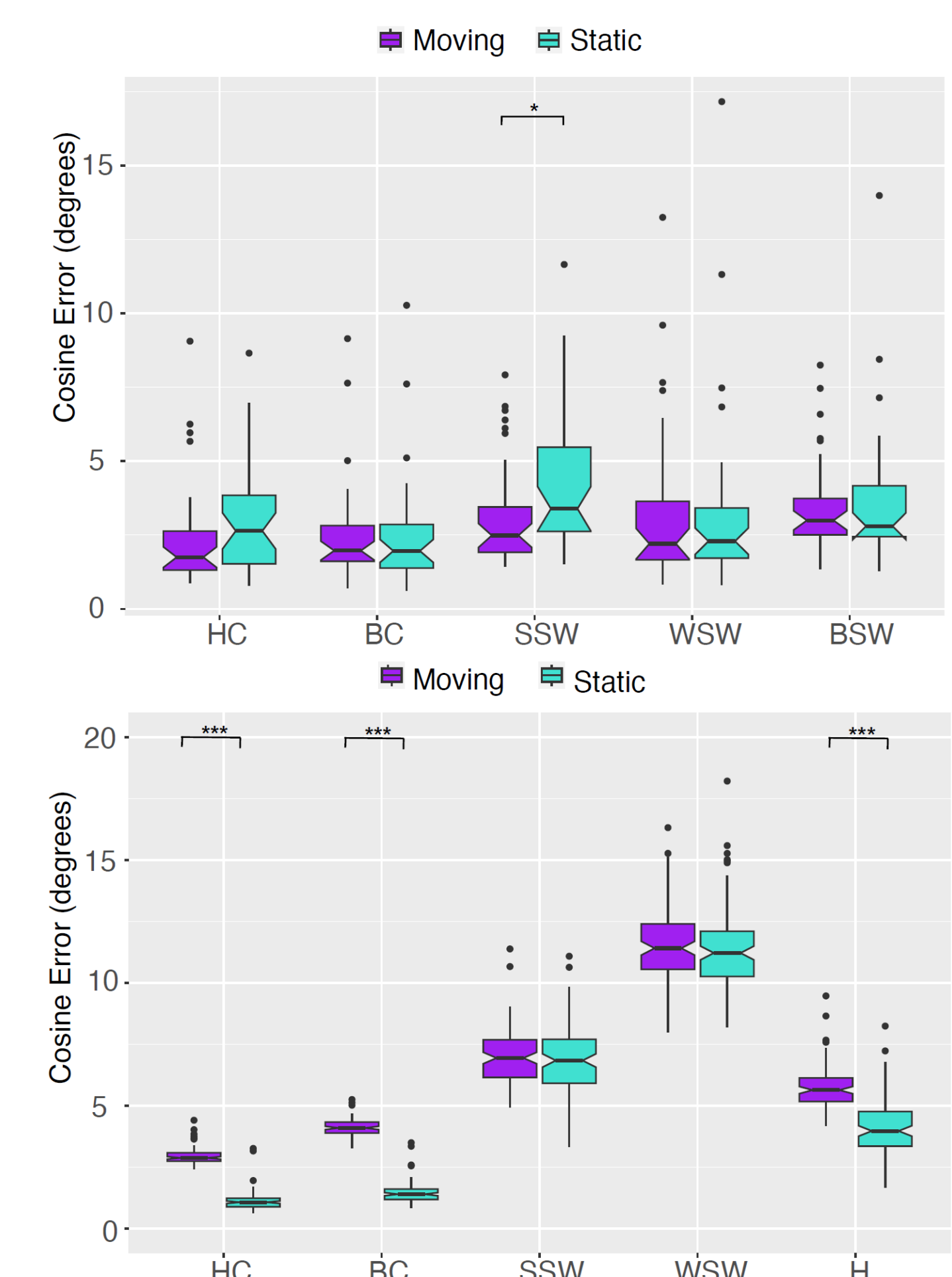
Recalibration

To see whether we could improve on our results with further recalibration on the other tasks, we used linear regression:

$$x' = A_x x + B_x y + C_x, \quad y' = A_y x + B_y y + C_y$$

The coefficients A , B , and C were calculated using the data from whichever task was used for recalibration and the coefficients were applied to data from other tasks.

STIMULUS MOVEMENT



- The HoloLens 2 data exhibited a clear trend
- Moving stimulus resulted in lower accuracy compared to the static case.

Figure 5 Cosine error for the Magic Leap (top) and HoloLens 2 (bottom), separated by stimulus movement. Notches show 95% confidence interval, * denote $p < 0.05$, and *** denote $p < 0.01$.

We conducted two separate user studies on the devices Magic Leap (n=36, 1 trial per task) and HoloLens 2 (n=54, 2 trials per task). We outline the tasks presented in the two studies below:

R Users rest their head on a chin rest and watch a tracking stimulus on the screen, which shifts to random positions after a brief delay.

HC Users rest their head on a chin rest and watch a tracking stimulus, which moves along a path on the screen inside their field of view (FOV).

BC Users sit and turn their heads to watch a tracking stimulus, which moves along a path in the world FOR and wider than their FOV.

SSW Users walk in circles around a table while watching a tracking stimulus, which moves along a path on the screen inside their FOV.

WSW Users walk in circles around a table while watching a tracking stimulus, which moves along a path in the world FOR above the table.

BSW Users walk in circles around a table while watching a tracking stimulus, which moves along a path constrained to a controller strapped to their body.

H Users walk down a hallway while watching a tracking stimulus, which moves down the hallway in front of them.

RECALIBRATION

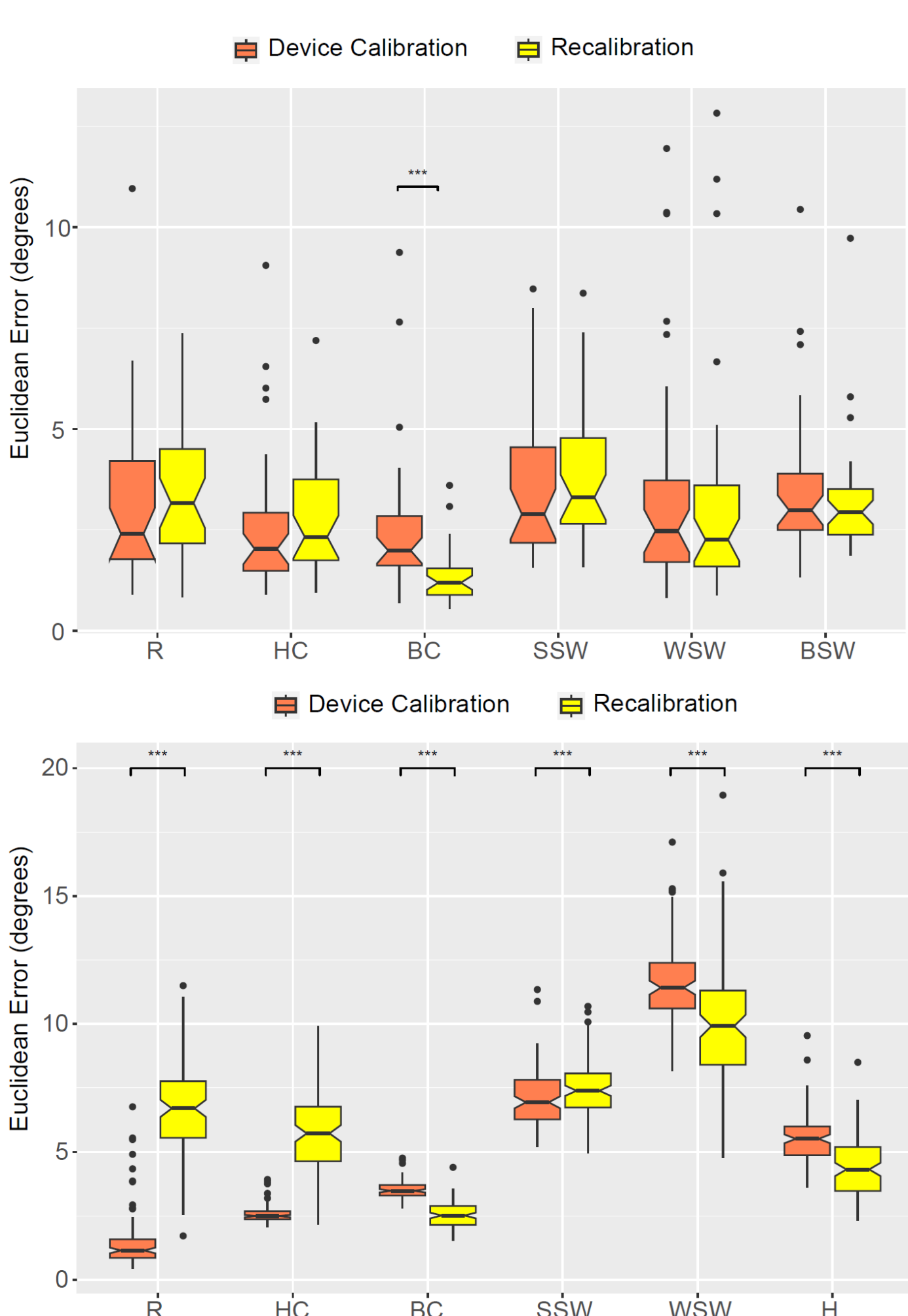


Figure 3 Euclidean error graphs for Magic Leap and HoloLens 2, recalibrated on the BC task. Notches show 95% confidence interval, *** denote $p < 0.001$.

- Recalibration using higher movement tasks such as the BC improves locomotion conditions
- Results in worse performance during the head-constrained tasks
- Worthwhile to consider the end use case of a headset when choosing a recalibration task

DEVICE COMPARISON

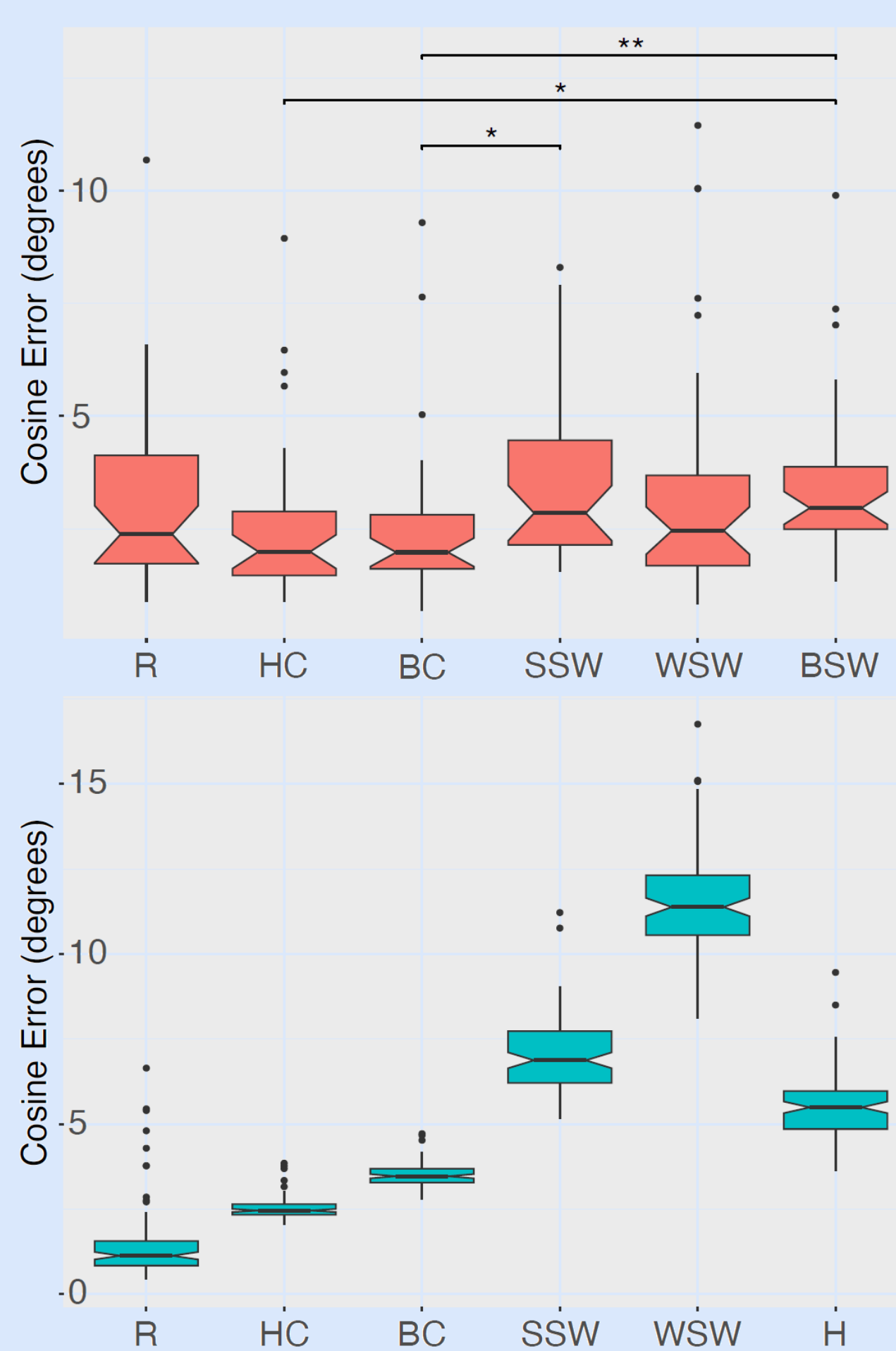


Figure 4 Cosine error per task for the Magic Leap (top) and HoloLens 2 (bottom). Notches show 95% confidence interval, * denotes $p < 0.05$, and ** denote $p < 0.01$. Significance bars for bottom omitted as all pairs differ significantly from each other.

- Magic Leap data displayed a generally higher accuracy, it is a more stable fit
- Very limited loss of accuracy due to locomotion or head motion in it.
- Wobble in HoloLens 2 headset could be why the it is more accurate only when the head is kept still, compared to the Magic Leap

REFERENCES

- [1] B. T. Carter and S. G. Luke. Best practices in eye tracking research. International Journal of Psychophysiology, 155:49–62, 2020. doi: 10.1016/j.ijpsycho.2020.05.010
- [2] A. T. Duchowski. Eye tracking methodology: Theory and practice. Springer, 2017
- [3] C. Weigle and D. Banks. Analysis of eye-tracking experiments performed on a tobii t60. 01 2008. doi: 10.1117/12.768424
- [4] B. V. Ehinger, K. Groß, I. Ibs, and P. K'önig. A new comprehensive eye-tracking test battery concurrently evaluating the pupil labs glasses and the EyeLink 1000. PeerJ, 7(e7086):e7086, July 2019.
- [5] S. Aziz and O. Komogortsev. An assessment of the eye tracking signal quality captured in the HoloLens 2. In 2022 Symposium on Eye Tracking Research and Applications. ACM, jun 2022. doi: 10.1145/3517031.3529626
- [6] A. T. Duchowski. A breadth-first survey of eye-tracking applications. Behav. Res. Methods Instrum. Comput., 34(4):455–470, Nov. 2002. doi:10.3758/BF03195475
- [7] A. Poole and L. Ball. Eye tracking in human-computer interaction and usability research: Current status and future prospects, pp. 211–219.01 2006.